OPLS All-Atom Force Field

The parameters for the force field are distributed with the BOSS and MCPRO programs in the files *oplsaa.par* and *oplsaa.sb*.

Form of the Force Field

Bond stretching:

$$E_{bond} = \sum_{bonds} K_r (r - r_{eq})^2$$

Angle bending:

$$E_{angle} = \sum_{angles} K_\theta (\theta - \theta_{eq})^2$$

Torsion:

$$E_{dih} = V_1(1 + \cos\phi)/2 + V_2(1 - \cos 2\phi)/2 + V_3(1 + \cos 3\phi)/2 + V_4(1 - \cos 4\phi)/2$$

Non-bonded:

$$E_{ab} = \sum_{i}^{on\ a} \sum_{j}^{on\ b} [q_i q_j e^2 / r_{ij} + 4\varepsilon_{ij}(\sigma_{ij}^{12} / r_{ij}^{12} - \sigma_{ij}^{6} / r_{ij}^{6})] f_{ij}$$

f_{ij} = 0.5 if i, j are 1,4; otherwise, f_{ij} = 1.0

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OPLS-AA atom types for bond stretching, angle bending and torsions.
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AMBER all-atom definitions are used (JACS 117, 5179 (1995), Table 1)
with the following additions:

CO        acetal C  (O- CR2 -O)
C=        C2 in dienes (C1 is CM)
CZ        sp C in triple bond
C+        carbenium carbon
C!        C1 in biphenyl-like ring
CS        C2 in heterocycle 5-ring like pyrrole, furan, thiophene
CU        C next to N:-X in aromatic 5-ring like pyrazole, isoxazole
CX        C in C=C in protonated imidazole, histidine
CY        sp3 carbon in 3-ring or 4-ring
C$        carbonyl carbon in 3-ring or 4-ring (e.g, lactam)
C:        allene or ketene =C=

N$        amide N in 3-ring or 4-ring
NZ        sp N in triple bond or azide
NT        amine, aniline N
NO        N in N=O (nitro, nitroso)
N=        N2 in azadiene C=N-C=C, etc.
NM        N in tertiary amide
from AMBER: N, NA, NB, NC, N2, N3, NY, N*

ON        O in N=O
OY        O in S=O
O$        O in 3-ring or 4-ring, e.g., epoxide

P+        phosphonium P

SY        S in S=O in sulfonamide or sulfone
SZ        S in sulfoxide
S=        S in thiocarbonyl (thiourea, thioamide,...)
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Summary of OPLS-AA Atom Types for Amino Acids

Backbone entries are:

Atom	Type
amide C	235
amide O	236
Calpha in Gly	223
Calpha in Pro	246
Calpha in Ala, etc.	224
Calpha in Aib	225
H-Calpha in all AA	140
N sec-amide; all AA except Pro	238
N tert-amide; Pro	239
H-N in all AA	241

Side-chain entries are:
H on any saturated C is type 140
H on any sp2 C is type 146

AA	Atom	Type	AA	Atom	Type	AA	Atom	Type
Ala	CB	135	Aib	CB	135			
Val	CB	137	Val	CG	135			
Leu	CB	136	Leu	CG	137	Leu	CD	135
Ile	CB	137	Ile	CG1	135	Ile	CG2	136
Ile	CD2	135						
Pro	CB	136	Pro	CG	136	Pro	CD	245
Phe	CB	149	Phe	CG-CZ	145			
Ser	CB	157	Ser	OG	154	Ser	HO	155
Thr	CB	158	Thr	OG	154	Thr	HO	155
Thr	CG	135						
Tyr	CB	149	Tyr	CG-CE	145	Tyr	CZ	166
Tyr	O	167	Tyr	HO	168			
Trp	CB	136	Trp	CG	500	Trp	CD (CH)	145
Trp	CD (C)	501	Trp	NE	503	Trp	HN	504
Trp	CE (C)	502	Trp	other CH	145			
Cys	CB	206	Cys	SG	200	Cys	HS	204
Cyx	CB	214	Cyx	SG	203			
Met	CB	136	Met	CG	210	Met	SD	202
Met	CE	209						
Asp	CB	274	Asp	CG	271	Asp	OD	272
Glu	CB	136	Glu	CG	274	Glu	CD	271
Glu	OE	272						
Asn	CB	136	Asn	CG	235	Asn	OD	236
Asn	ND	237	Asn	HN	240			
Gln	CB	136	Gln	CG	136	Gln	CD	235
Gln	OE	236	Gln	NE	237	Gln	HN	240
Lys	CB-CD	136	Lys	CE	292	Lys	NZ	287
Lys	HN	290						
Arg	CB	136	Arg	CG	308	Arg	CD	307
Arg	NE	303	Arg	HNE	304	Arg	CZ	302
Arg	NN	300	Arg	HNN	301			
Hid	CB	505	Hid	CG	508	Hid	ND1	503
Hid	HND	504	Hid	CE1	506	Hid	HE	146
Hid	NE2	511	Hid	CD2	507	Hid	HD	146
Hie	CB	505	Hie	CG	507	Hie	ND1	511
Hie	CE1	506	Hie	HE	146	Hie	NE2	503
Hie	HNE	504	Hie	CD2	508	Hie	HD	146

Hip	CB	505	Hip	CG	510	Hip	ND1	512
Hip	HND	513	Hip	CE1	509	Hip	HE	146
Hip	NE2	512	Hip	HNE	513	Hip	CD2	510
Hip	HD	146						